

Khanjan Jha

Full-Stack Engineer · React · Node.js · AI/RAG · Three.js

jhakhanjan62@gmail.com 9990871334 [GitHub](#) [LinkedIn](#) [Portfolio](#)

Full-Stack Software Engineer with 1+ years of experience building scalable web applications using React, Next.js, Node.js, and TypeScript. Proven track record of shipping production-grade features, reducing load times by up to 50%, and integrating AI/LLM-powered workflows. Passionate about 3D graphics, RAG pipelines, and clean, maintainable architecture.

EXPERIENCE

Software Engineer — Freelancer

Jan 2026 – Present

Remote

- Architected and shipped full-stack features across 8+ client projects using React, Next.js, Node.js, and MongoDB.
- Integrated third-party REST APIs with robust error boundaries, retry logic, and loading state management.
- Optimized rendering performance for graphics-intensive applications, reducing frame drops by 25% through GPU-accelerated post-processing and progressive asset streaming.
- Maintained and debugged web applications for performance and crossbrowser compatibility, ensuring consistent behavior across Chrome, Firefox, and Safari.

Software Development Intern — CARS24 (Product Team)

Feb 2025 – Nov 2025

Gurugram, IN

- Built and maintained production-grade React + TypeScript applications serving internal operations across 200+ cities.
- Created **GridBuilder**, a config-driven reusable data-grid component that cut feature development time by 40%.
- Collaborated with backend engineers on REST API contract design, reducing frontend/backend integration bugs.
- Led performance upgrades for internal admin panels, translating feedback from cross-functional teams (Support, Admins) into high-usability features and optimized rendering workflows.
- Engineered interactive 3D visualizations for Orbit (Cars24) using Three.js/React Three Fiber, optimizing GLTF assets and HDRI lighting to boost engagement to 18k monthly visitors

Full-Stack Developer Intern — ALTIE Reality

May 2024 – Jul 2024

Remote

- Built **LearnXR**, a responsive 3D learning platform (React.js + GSAP + Three.js) with optimized model loading for cross-device performance.
- Engineered **IN3D.Ai**, a WebGL-powered 3D web app with Google OAuth, JWT auth, and real-time backend synchronization.
- Reduced 3D asset load times by 35% using GLTF compression and progressive texture streaming techniques.

PROJECTS

Multi Tuning ChatBot [GitHub](#) [Link](#)

- Architected and developed a multi-tenant AI chatbot platform using Node.js and MongoDB, supporting isolated workspaces, role-based access control (RBAC), custom knowledge bases, and organization-level data segregation.
- Built an end-to-end RAG pipeline with PDF ingestion, intelligent chunking, Google embeddings, MongoDB Atlas Vector Search, and Groq-powered LLM inference, enabling context-aware document question answering.
- Reduced token consumption and inference costs by implementing context engineering techniques including conversation summarization, memory compression, retrieval optimization, and dynamic context window management.
- Designed scalable backend infrastructure with JWT authentication, real-time SSE streaming, asynchronous embedding-generation workers, queue-based document processing, conversation persistence, and chatbot configuration APIs.

3D Car Simulator [Link](#)

- Developed a real-time 3D car configurator (React Three Fiber + WebGL) supporting 10+ model/environment combinations with live state switching.
- Implemented GPU-optimized post-processing (Bloom, SSR, SSAO, Tone Mapping) while sustaining stable 60 FPS on mid-range hardware.
- Reduced initial load time 45% via GLTF Draco compression, LOD meshes, and progressive asset streaming.

Movie Discovery App [Link](#)

- Built a full-featured movie discovery app with TMDB API, Redux Toolkit, and infinite scrolling for a seamless browsing experience.
- Reduced redundant API calls by 35% with RTK Query caching; debounced search with 300ms delay for real-time results.
- Designed and optimized infinite scrolling and debounced advanced search, allowing users to explore thousands of movies without performance drops.

TECHNICAL SKILLS

Frontend	React.js, Next.js, TypeScript, JavaScript (ES6+), Tailwind CSS, Redux Toolkit, HTML5, CSS3
Backend	Node.js, Express.js, MongoDB, REST APIs, GraphQL, JWT/OAuth, WebSockets, SSE
AI / RAG	RAG Pipelines, Vector Search, LLM Integration (Groq, OLLAMA), Prompt Engineering, Embeddings
3D Graphics	Three.js, React Three Fiber, WebGL, GLSL, GLTF Optimization, Post-Processing
DevOps / Tools	Docker, Git, GitHub Actions, Vercel, Webpack

EDUCATION

Bachelor of Technology — Information Technology

2021 – 2025

Rajasthan Technical University, Kota